Applying the operational support systems architecture

S Pyzer, P Williams and D Richards

There are currently of the order of a thousand computer systems supporting BT's operational processes. This paper describes how the operational support systems architecture is being used to help rationalise and reduce this portfolio into a more manageable set by ensuring maximum reuse of functionality and minimum replication of data. This will lead to lower maintenance and development costs as well as speedier delivery of new solutions.

1. Introduction

The reasons for the development and the method adopted in the production of the BT architecture for the management of operational support systems (OSS) have been expounded elsewhere [1]. This paper reviews the salient parts of the architecture, describes the value of it and how BT uses it in the design, development, procurement and deployment of computer systems.

1.1 Background

The BT OSS architecture (see Fig 1) is just one of the components of the BT OSS framework [2].

The current scope of the architecture comprises three main areas:

- selling and billing of services,
- maintenance of service quality,
- and the management of the BT portfolio of services and its network platforms.

The architecture evolves as the requirements of the business change, as functionality is enhanced and as further detail is added. Work is currently in hand to increase the

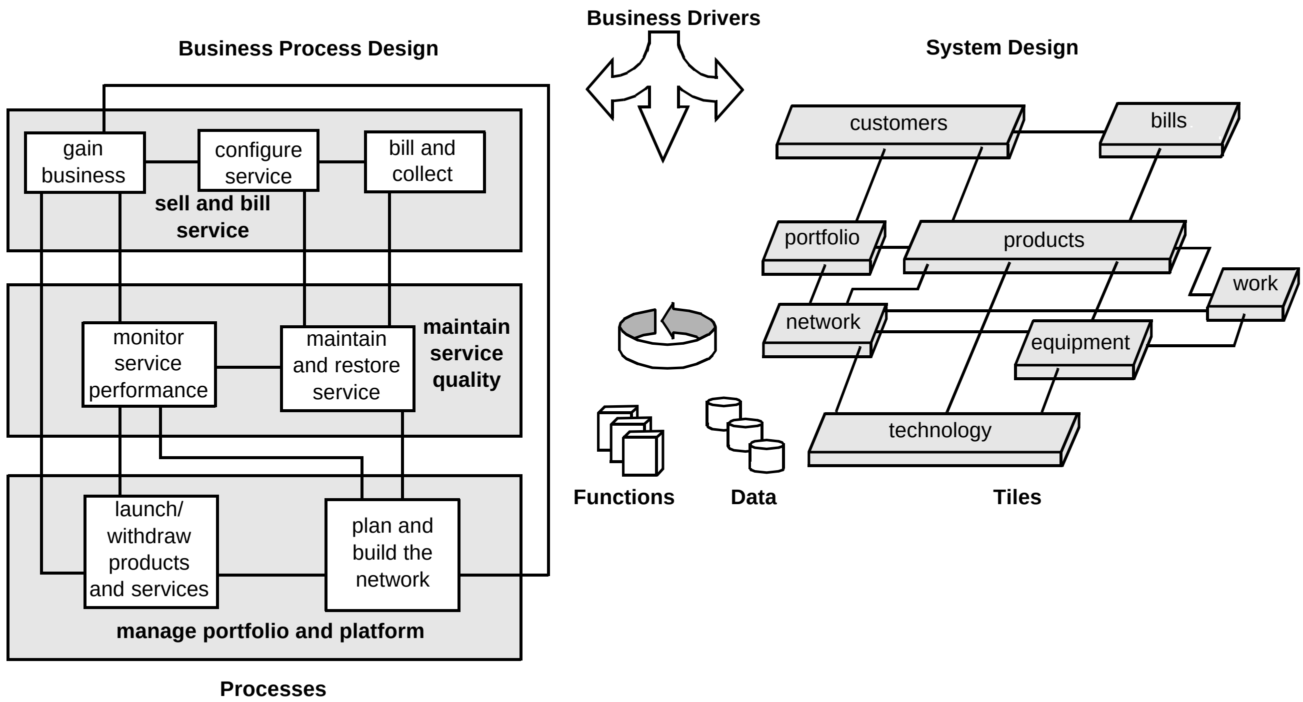

Fig 1 The BT OSS architecture.

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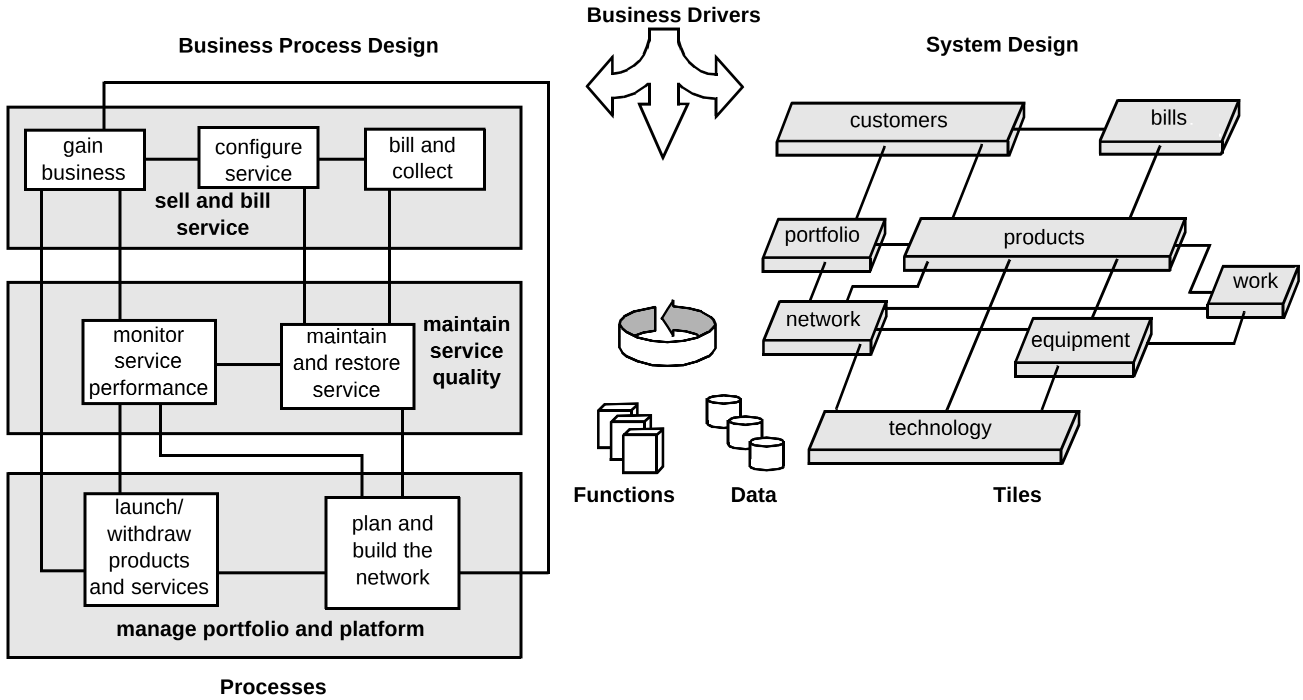

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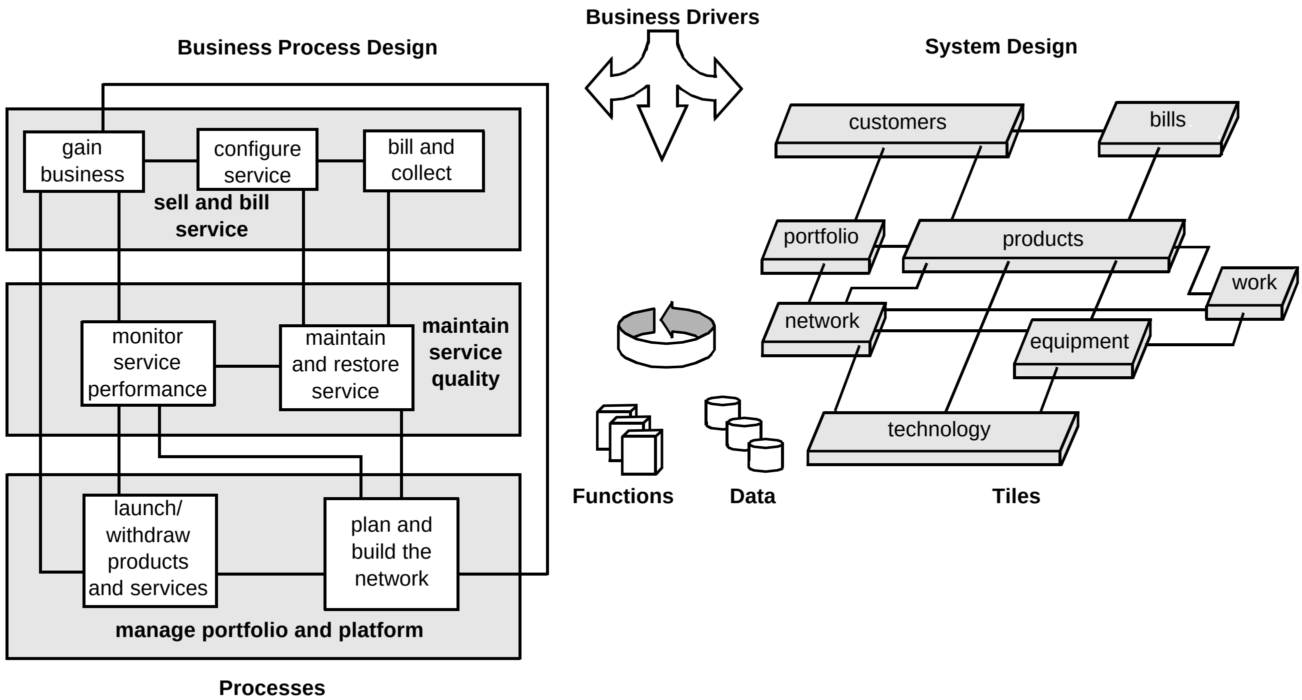


Fig 1 The BT OSS architecture.

scope to encompass other business areas of BT and its activities outside the UK.

The complete OSS framework consists of the following components:

- the BT OSS architecture [1],
- the data architecture, comprising data areas, data models, data flows, data standards, the interfaces between data areas and key business drivers — this is encapsulated in what is known as the data architecture manual (DAM),
- the portfolio of BT's computer systems — details of these are held in the BT systems encyclopaedia (BTSE),
- system evolution plans — these consist of a set of plans for the further development of each of BT's key systems over the next five years,
- reference solution designs — these are a set of designs that collectively provide up to 80% of the solution for new products. They are based on BT's key systems and are compiled from a set of tables of preferred software,
- solution designs — produced to meet specific requirements.

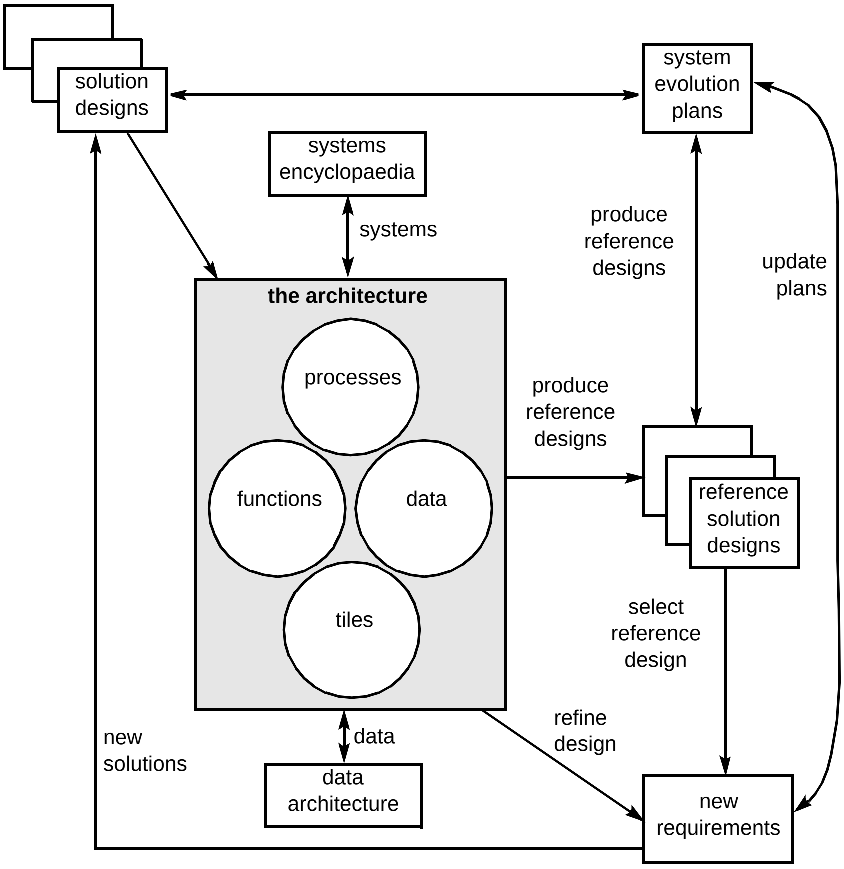
Figure 2 demonstrates how these various components are related and work together to provide the complete framework within which OSS developments are managed.

1.2 Context

It is the purpose of computer systems to serve the needs of the business and to respond to the pressures and drivers on the organisation. Computer systems need to automate processes, provide accurate information in time and be able to support the mix of products in order to deliver timely customer solutions both in the UK and globally. Also, at a time of change in the telecommunications market, the computer systems must be flexible, not constrained to any particular organisational model and support the need for continuous improvement.

BT has a portfolio of computer systems which must be managed. These systems support the business processes by encompassing the required functionality, appropriate data and data access. This portfolio depends on a technical architecture where data is well managed, secure and accurate, withone master source, and where interfaces are clearly defined, effective and managed.

The OSS architecture is a structure which aids understanding of the roles and relationships of the computer systems. It is required because of the sheer complexity of



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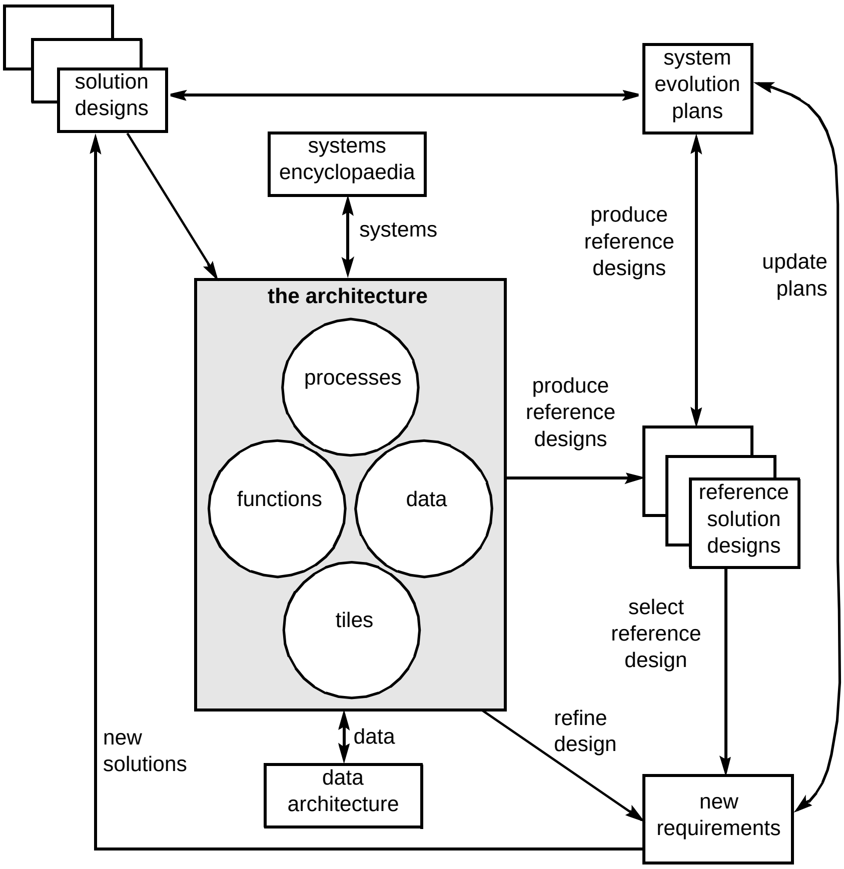

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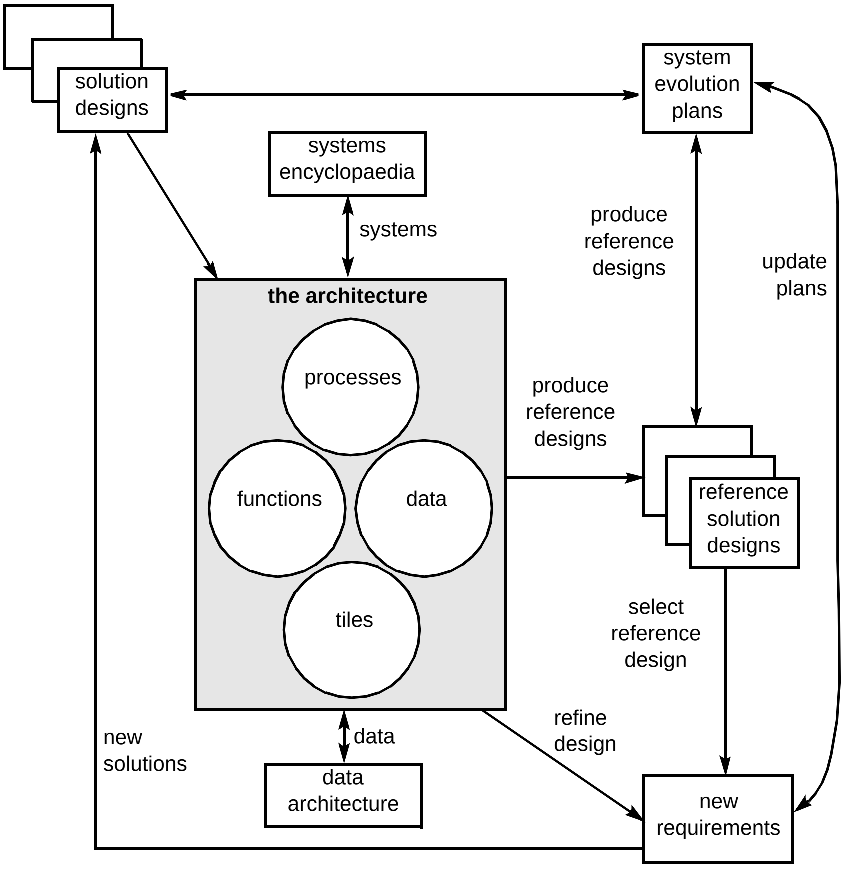

Fig 2 Components of the OSS framework.

modern telecommunications and information networks and the number and scale of BT's systems.

The architecture will help the company by:

- improving the quality of software development,
- improving data management capabilities — availability, accuracy, and controlled replication,
- increasing the speed of systems development,
- assisting in the handling, assessment and delivery of required changes,
- allowing for greater management control of the portfolio,
- ensuring reusability becomes a reality,
- defining and identifying ownership.

The architecture also acts as an enabler for the promulgation of standards and data quality.

1.3 Structure of the BT OSS architecture

The architecture has the following parts — generic business processes, component processes, actions, data, functions, tiles, and all their associated interfaces. Figure 3 shows the inter-relationships of these parts of the architecture and the relationships with their physical implementations.

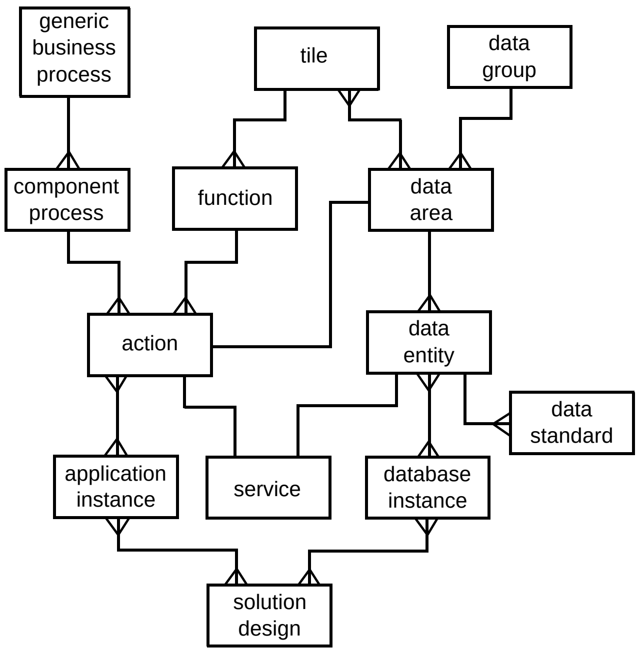

Fig 3 The framework information relationships.

Generic business processes

The framework has identified three key business processes which, in turn, have been broken down into seven generic business processes as follows:

- sell and bill service:

  — bill and collect,
  — configure service,
  — gain business.

- maintain service quality:

  — maintain and restore service,
  — monitor service performance,

- manage portfolio and platform

  — launch and withdraw products and services,
  — plan and build the network.

Component processes

Each generic business process has been broken down into a set of component processes which have been linked together with their required data. Each component process will perform an activity carried out by the company in the performance of its day-to-day operations and consists of all the actions needed to operate that process.

Actions

Both a function and a component process consist of actions. An action is in effect a lower level process and can be considered as the basic reusable building block for realising both component processes and functions. Based on the business context, they are logically grouped together to form component processes. In addition, based on affinity analysis for their requirement for access to like items of data, they are also logically grouped together to form functions. This gives the architecture its great power of linking the logical world (the processes) with the physical world (functionality as delivered by systems).

Data

This is defined in terms of data areas which are logical groupings of closely related data items. The data areas describe things about which the company is interested, and the ways in which they are behaving.

Functions

At the logical level, a function is a grouping of actions based on their affinity for like items of data. At the physical level, it is considered to be a suitable candidate for automation wholly within one application.

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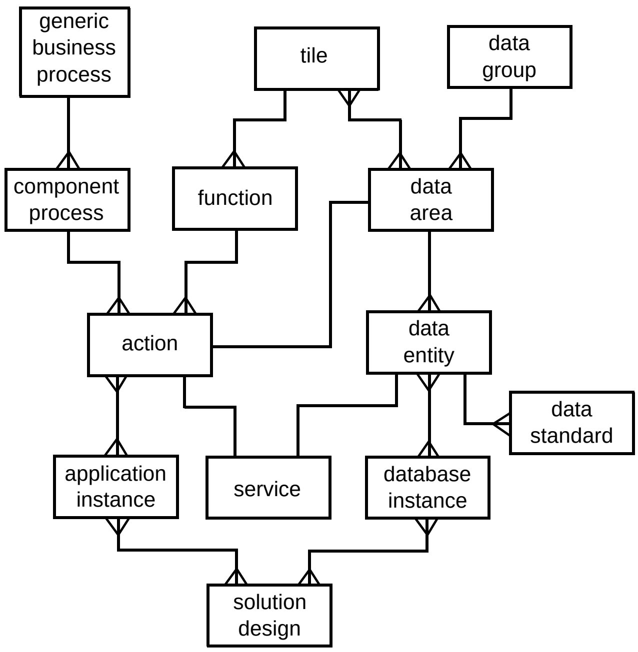

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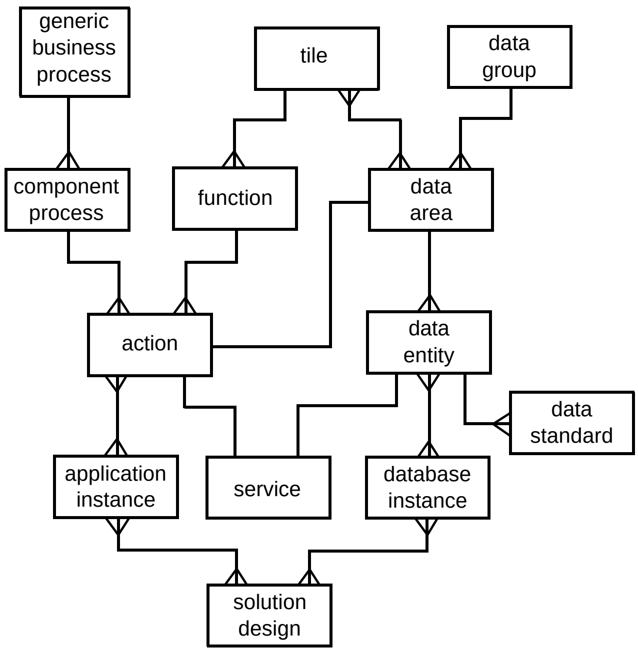

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2. Value of the architecture

A very major role played by the architecture is the tight linkage it provides between the logical process world and the physical world of people and computer systems.

It provides a structure for the effective management of data and computer systems to serve BT's large and complex business. Use of the architecture enables consistency within the designs which take a company-wide view, rather than being restricted to a particular product, service, process, technology or geographical area. It also makes it clear where a focused design fits in the broader picture, exposing the relationships and dependencies.

The advantages to the company of using the architecture include:

- a common understanding which is gained through using standard process and function design,
- consistency of presentation of the computer systems solutions,
- the enabling of ownership for data and functions,
- the clear definition of the scope of solutions and associated responsibilities,
- the provision of a common language or terminology to encourage consistency in understanding and expression of designs.

The architecture also gives a structure in which issues can be examined, e.g. whether software solutions should be acquired from external suppliers as opposed to being developed in-house and whether a proposed solution for one business requirement is consistent and fitting compared to existing applications and proposed solutions in other areas. When using the same structure for all designs, the probability of reusing the same or existing components for many solutions will increase.

The framework allows BT to model the operations of other administrations, other licensed operators and service providers without being concerned with the detail of their operational processes and computer systems.

3. How the architecture helps the design process

The role that the architecture plays in the delivery of systems is to assist in the production of solution designs and evolution plans. Solution designs need to be produced to support new business objectives and initiatives. Evolution plans need to be developed for existing systems.

For existing systems, the architecture assists in the analysis by characterising these systems in terms of the functions they perform and the data which they hold or use. This will enable designers to place the system on the appropriate tile(s). The overall design policy which BT is following is to concentrate design and development effort on key systems with clearly defined and delimited functionality. It is intended that the functionality provided by any one system should be wholly contained within one tile. As existing systems, by definition, were designed and developed prior to the introduction of the architecture, it is not necessarily expected that they will align with this view and it is often found that an existing system has an appearance on more than one tile.

For new applications, the architecture will help solution designers to ensure consistency between solutions. Its scope covers the whole of service and network management and thus enables designers to produce designs for specific requirements with a knowledge of the broader context. A checklist of all the issues which need to be taken into consideration in producing a design is also included in the architecture.

Design takes place within the context of key systems. All current key systems have been analysed for the functionality delivered. This functionality has been considered from different business perspectives, such as technology (e.g. ISDN), geography (e.g. Pacific Rim, UK), products and services (e.g. FeatureNet, PSTN). Thus a complete picture has been built up of what each of the key systems deliver within various categories.

3.1 Guidelines/principles

A basic set of principles has been created to guide the design of new systems and the evolution planning of current systems. The architecture provides valuable input and further guidance in the implementation of these principles. The following paragraphs give the authors' views as to how the architecture will contribute to this process.

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- Solutions should reduce the number of manual interventions required in the company's processes, thereby reducing the cost of failure — the close linkage of the logical and physical views in the architecture provide designers with a clear view of where there are gaps in automation.

- Modular solutions will be built with clear boundaries and interfaces — the architecture provides very clearly scoped and unambiguously defined functions, together with their interfaces. These modular solutions will be the physical implementation of those functions.

- The model for system deployment is the three-tier architecture — one of the major criteria behind the development of the architecture has been to keep the functionality and the data separate. Therefore the architecture fully supports the three-tier requirement of user access, applications and data layers (see section 5).

- New system developments will only be instigated if the need cannot be met by an existing key system — the architecture includes a mapping of all key systems to the functionality supported by them, together with information on the business perspective. Reference to this will give a quick guide as to the areas where a current system has the potential of meeting the requirements.

- The scope and complexity (in terms of functionality and data) of existing systems will not be increased — the architecture is playing a major role in the scoping of systems. Evolution plans have been created for all of BT's key systems. Based on the architecture, these rationalise the functionality provided by each system such that, ultimately, no system provides functionality spanning more than one tile.

- All new functional interfaces will be based on TMN (telecommunications management network) standards where appropriate — this guideline, together with the tight and clear scoping of functionality of the modules as defined within the architecture, enhances the opportunities to buy in components.

- Key BT reference databases will be used directly as sources of data. Data replication will be minimised and fully controlled, and data should be obtained directly from identified master sources of data. If performance considerations require the use of copies of such data, the replication must be under strict control — the architecture clearly defines the data access requirements of the individual functions. This identifies all functionality that requires direct access to the master sources of data and others that can be provided by a controlled copy (for read access only).

4. Procurement

Single components or whole computer systems may be bought in from suppliers external to the company. These systems must then be integrated with other systems in terms of both functionality and data. The architecture is a useful tool for checking that a purchased solution meets the functional and data requirements of the business.

The procurement process should use any key business drivers identified within the architecture as a checklist in determining whether a package meets all or part of the requirements.

5. Deployment

The framework is not prescriptive concerning the deployment of applications and systems on a particular kind of hardware architecture. However, BT has adopted a three-tier model for its computer systems, known as the three-tier architecture.

The term 'three-tier architecture' may be used to classify the logic within an application — a logical three-tier architecture — and is also often applied to three physical classes of machines (workstations, mid-range servers and mainframe systems) — a physical three-tier architecture.

The OSS architecture is a logical framework within which computer systems are to be developed. It does not make any recommendations as to the exact physical architecture to be employed. However, it can be seen to fit in very neatly with this three-tier approach. Figure 4 is a diagrammatic representation of the three-tier architecture, with the tile architecture overlaid. The OSS architecture has not concerned itself with the presentation layer, but it can be seen that the tile architecture spans the mid-tier layer (the reusable functions representing the business logic) and the data layer (the data areas representing the physical databases to be employed).

6. Rationalisation

Aside from the design and development of solutions, the framework is used as a base for the creation of evolution plans with a view to rationalising BT's current systems into a more manageable set. Overall, the approach to rationalisation is to:

- partition the existing systems into two categories — key and non-key systems,

- Solutions should reduce the number of manual interventions required in the company's processes, thereby reducing the cost of failure — the close linkage of the logical and physical views in the architecture provide designers with a clear view of where there are gaps in automation.

- Modular solutions will be built with clear boundaries and interfaces — the architecture provides very clearly scoped and unambiguously defined functions, together with their interfaces. These modular solutions will be the physical implementation of those functions.

- The model for system deployment is the three-tier architecture — one of the major criteria behind the development of the architecture has been to keep the functionality and the data separate. Therefore the architecture fully supports the three-tier requirement of user access, applications and data layers (see section 5).

- New system developments will only be instigated if the need cannot be met by an existing key system — the architecture includes a mapping of all key systems to the functionality supported by them, together with information on the business perspective. Reference to this will give a quick guide as to the areas where a current system has the potential of meeting the requirements.

- The scope and complexity (in terms of functionality and data) of existing systems will not be increased — the architecture is playing a major role in the scoping of systems. Evolution plans have been created for all of BT's key systems. Based on the architecture, these rationalise the functionality provided by each system such that, ultimately, no system provides functionality spanning more than one tile.

- All new functional interfaces will be based on TMN (telecommunications management network) standards where appropriate — this guideline, together with the tight and clear scoping of functionality of the modules as defined within the architecture, enhances the opportunities to buy in components.

- Key BT reference databases will be used directly as sources of data. Data replication will be minimised and fully controlled, and data should be obtained directly from identified master sources of data. If performance considerations require the use of copies of such data, the replication must be under strict control — the architecture clearly defines the data access requirements of the individual functions. This identifies all functionality that requires direct access to the master sources of data and others that can be provided by a controlled copy (for read access only).

4. Procurement

Single components or whole computer systems may be bought in from suppliers external to the company. These systems must then be integrated with other systems in terms of both functionality and data. The architecture is a useful tool for checking that a purchased solution meets the functional and data requirements of the business.

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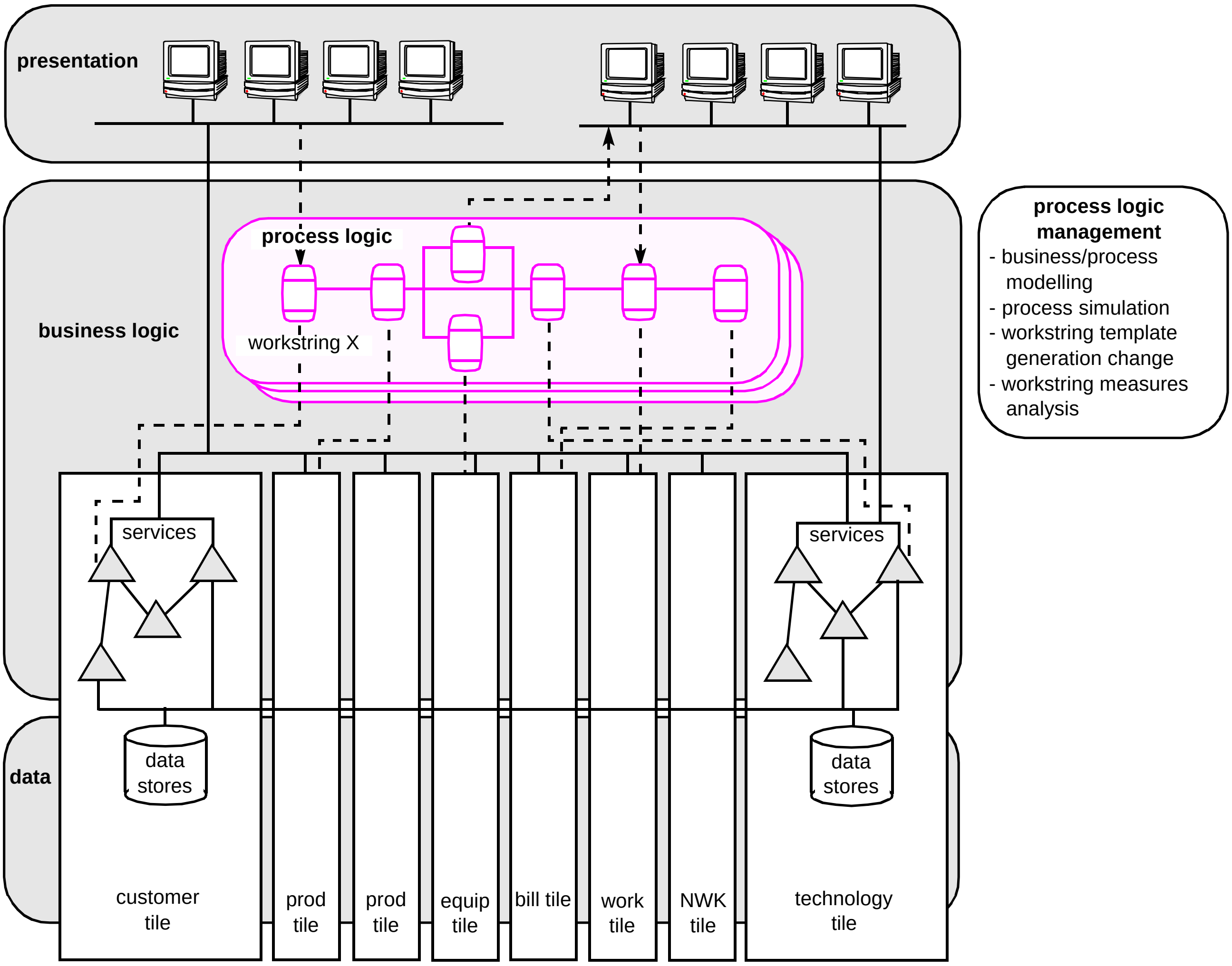


Fig 4 Relationship of the three-tier architecture with the OSS architecture.

- taking into account each product and service, produce evolution plans for key systems which support the business processes,

- identify candidate systems where there is scope for rationalisation because of data and functional duplication.

The framework provides the process, data and function structure which enables the above analysis.

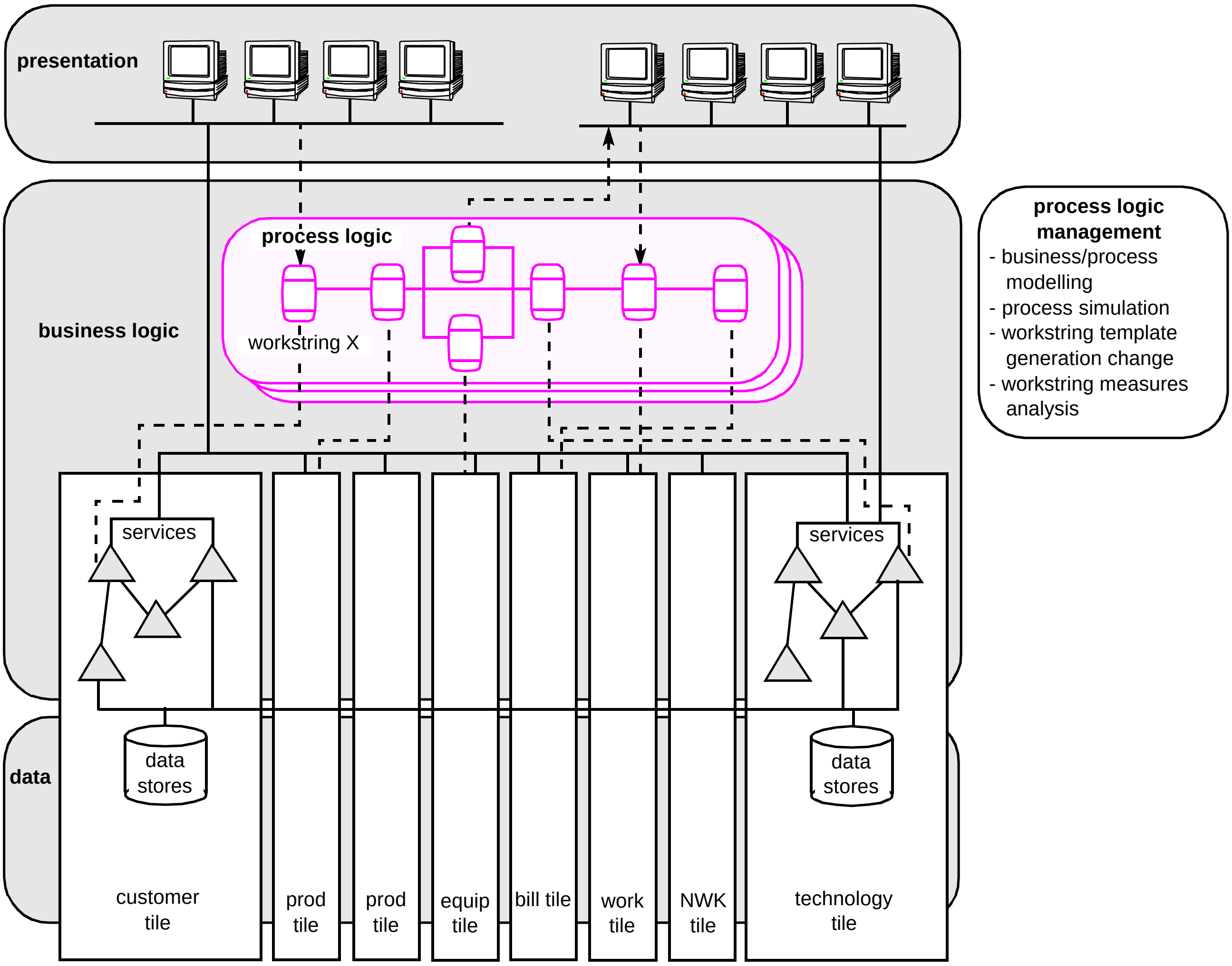
Key systems are those which form the core of the company's computing power. They are the systems which support the main processes, tend to be the larger systems, and the ones on which most of the computer systems budget is spent.

Key systems are characterised in terms of the functionality and data held or used. When the key systems are considered as a set, it is possible to identify overlaps in functionality, having taken account of factors such as geographical coverage, network coverage, support of products and services, and network technology.

The changes over time to key systems are shown in the form of tile evolution plans. It is then possible to plot the evolution of the key systems and plan to rationalise them by bringing functionality together on these systems.

For non-key systems, a similar characterisation will be made of the functionality and data. Consideration can then be given as to whether key systems can supply the functionality and data manipulation required. Redundant and duplicate functionality and the transfer of functionality need to be considered on a case-by-case basis within the context of the product and service.

As functionality is moved from non-key systems, they can be gradually phased out.



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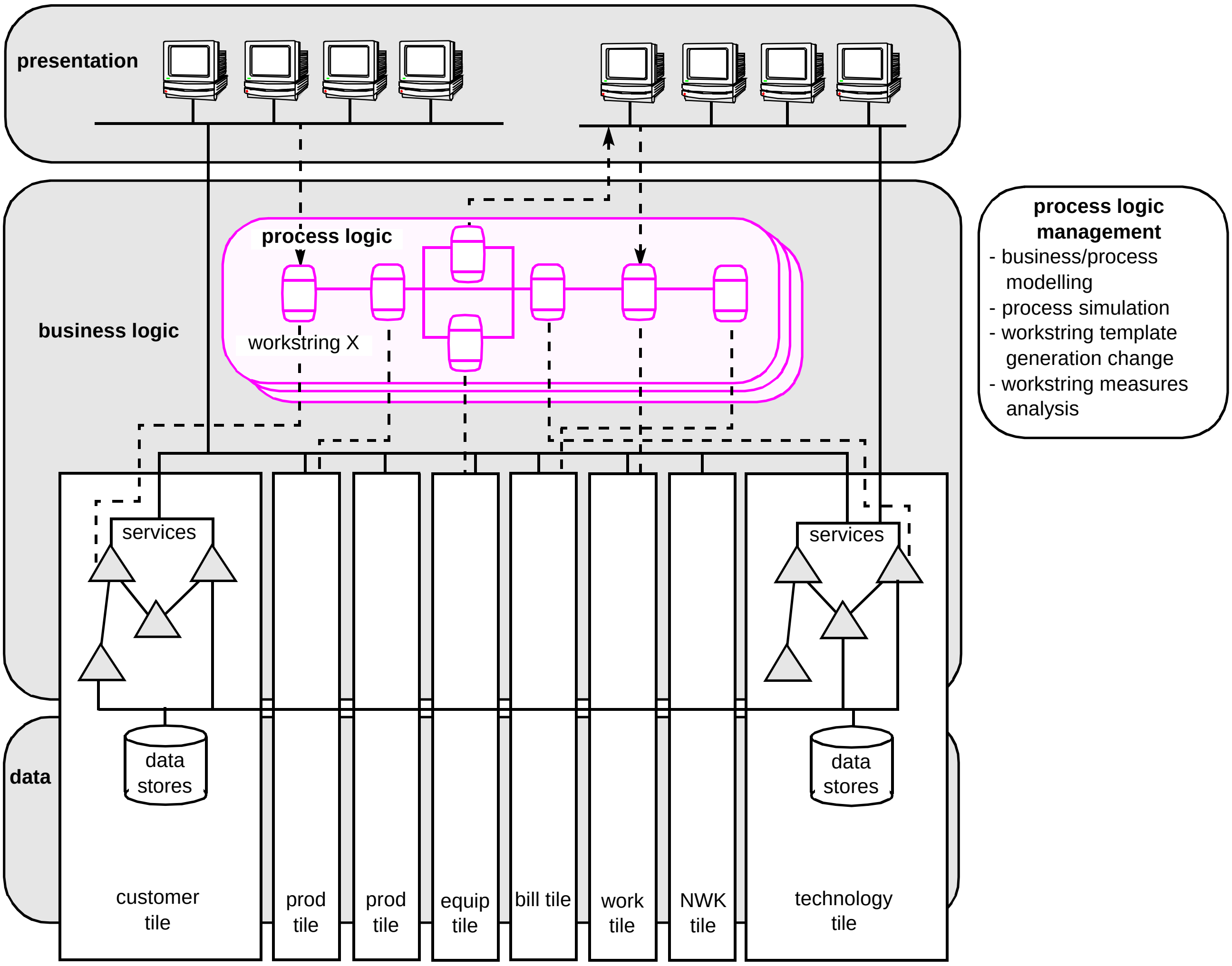

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7. Conclusions

This paper has described how the OSS architecture is helping BT in its quest to rationalise its many computer systems into a manageable quantity. It has also described how the architecture is part of a larger framework that provides the company with the means to manage future developments in a more controlled and flexible way. It is envisaged that use of the architecture will lead to:

- a reduction in the number of operational systems and their interfaces,
- a reduction in data replication,
- increased control of data,
- increased reuse of functionality,
- standardised interfaces,
- lower costs,
- quicker delivery,
- higher quality.

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